



Stony Brook Institute
at Anhui University

安徽大学纽约石溪学院

ISE 333: User Interface Development (Design)

Chapter 10: Devices

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Keyboards

- QWERTY Keyboard



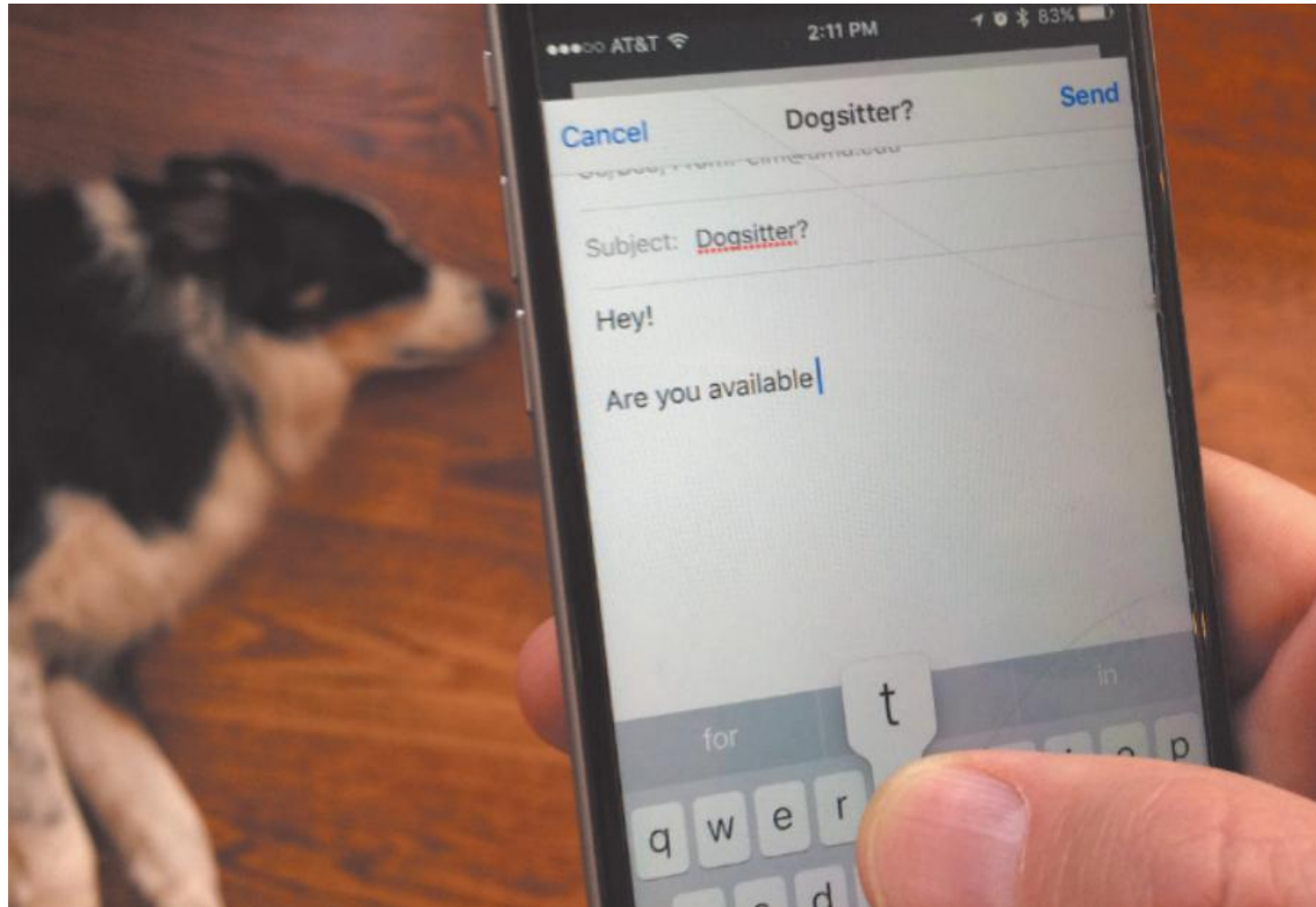
Keyboards

- Special Keyboards



Keyboards

- Special Keyboards



Pointing Devices

- Types of Interaction Tasks
 - Select
 - Choose from a set of items
 - Position
 - Choose a point in a one-, two-, or higher-dimensional space
 - Orient
 - Choose a direction in a two-, or higher-dimensional space
 - The direction may rotate a symbol on the screen, indicate a direction of motion, or control the operation of a device (say a robot arm)
 - Path
 - Define a series of positioning and orientation operations

Pointing Devices

- Types of Interaction Tasks

- Quantify

- Specify a numeric value

- Gesture

- Perform an action by executing a pre-defined motion
 - Dwelling on an object to bring up a context menu
 - Swiping to the left (or right) to turn a page forward (or backward)
 - Pinching (or separating) your fingers to zoom out (or in)

- Text

- Enter, move, and edit text in 2-D space
 - The pointing device indicates the location of an insertion, deletion, or change



Pointing Devices

- Types of Pointing Devices

Direct control devices (easy to learn and use, but hand may obscure display)

- Touchscreen (single- and multi-touch)
- Stylus (passive and active)

Indirect control devices (take time to learn)

- Mouse
- Trackball
- Joystick
- Pointing stick (trackpoint)
- Touchpad
- Graphics tablet

Novel devices and strategies (for special purposes)

- Bimanual input
- Eye-trackers
- Sensors (accelerometer, gyroscopes, depth cameras)
- 3-D trackers
- Data gloves
- Haptic feedback
- Foot controls
- Tangible user interfaces
- Digital paper

Pointing Devices

- Direct Control Devices
 - Stylus



Pointing Devices

- Indirect Control Devices
 - Mouse



Pointing Devices

- Indirect Control Devices
 - Joysticks



Pointing Devices

- Novel Control Devices
 - Hand tracker

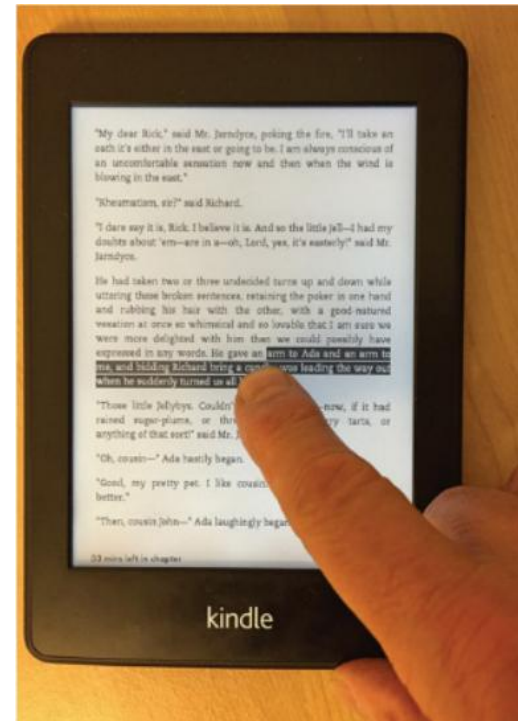


Displays

- Conception
 - The display is the primary source of visual feedback to users from UI
- Display Characteristics
 - Physical dimensions (usually the diagonal dimension)
 - Resolution (the number of pixels available)
 - Number of available colours and colour correctness
 - Luminance, contrast, and glare
 - Power consumption
 - Refresh rates (sufficient to allow animation and video)
 - Cost
 - Reliability

Displays

- Display Technology
 - Cathode-ray tube (CRT)
 - Liquid-crystal display (LCD)
 - Light-emitting diode (LED)
 - Organic light-emitting diode (OLED)



Displays

- Large Wall Displays



Displays

- Large Wall Displays

Users discussing and pointing at details on the Stony Brook University Reality Deck (Papadopoulos et al., 2015), an immersive gigapixel display consisting of 416 thin-bezel LCD displays and powered by 18 graphics workstations connected using a high-speed network (<https://labs.cs.sunysb.edu/labs/vislab/reality-deck-home/>).



Displays

- Large Wall Displays

Two users collaboratively control a lens on a gigapixel image of Paris, France, using a tablet touchscreen as well as an interactive cursor (Chapuis et al., 2014).
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Displays

- Tabletop (Horizontal) Displays



Displays

- Deformable (Shape-Changing) Displays

